We have seen that

$$\frac{d}{dx}(f+g)(x) = \frac{df}{dx}(x) + \frac{dg}{dx}(x) \qquad \text{(Sum Rule)}$$
$$\frac{d}{dx}(f-g)(x) = \frac{df}{dx}(x) - \frac{dg}{dx}(x) \qquad \text{(Difference Rule)}$$

1. (a) In light of the Sum and Difference Rules, make a conjecture for $\frac{d}{dx}(fg)(x)$.

   (b) Consider the functions $f(x) = x$ and $g(x) = x^2$. Does your conjecture for $\frac{d}{dx}(fg)(x)$ give the right answer?

 Name:_____

(c) Prove the *Product Rule*:

$$\frac{d}{dx}(fg)(x) = (\frac{d}{dx}f(x))g(x) + f(x)\frac{d}{dx}g(x),$$

or, using prime notation,

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x)$$

You will need to use the following fact: f is continuous at a if and only if $\lim_{h\to 0} f(a+h) = f(a)$. (This equivalent definition of continuity comes from substituting $h = x - a$ into $\lim_{x\to a} f(x) = f(a)$ and noting that $x \to a$ if and only if $h \to 0$.)

Hint: To help you get started, let $F(x) = f(x)g(x)$. Then

$$\begin{aligned}
\frac{d}{dx}F(x) &= \lim_{x\to 0}\frac{F(x+h)-F(x)}{h} \\
&= \lim_{x\to 0}\frac{f(x+h)g(x+h)-f(x)g(x)}{h} \\
&= \lim_{x\to 0}\frac{f(x+h)g(x+h)-f(x)g(x)+f(x+h)g(x)-f(x+h)g(x)}{h} \\
&=
\end{aligned}$$

2. Compute $f'(x)$ if $f(x) = (2x^3)(3x^4)$. Verify your answer using the power rule.

3. If $h(x) = xg(x)$ and it is known that $g(3) = 5$ and $g'(3) = 2$, find $h'(3)$.

4. Find the derivative of $f(x) = (1+2x^2)(x-x^2)$ in two ways: by using the Product Rule and by performing the multiplication first. Do your answers agree?

5. Differentiate $y = \sqrt[3]{t}(t^2 + t + t^{-1})$.

6. (a) If f, g, and h are differentiable, find $(fgh)'(x)$.

(b) If $f = g = h$, use your formula in part (a) to find $\frac{d}{dx}f(x)$.

(c) Use your answer to part (b) to differentiate $y = (x^4 + 3x^3 + 17x + 82)^3$. Do not simplify your answer.